

Chapter 1: Rise of the Quantum Machines: Fundamentals

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- Quantum Computers Explained – Limits of Human Technology (<https://youtu.be/JhHMJCUMq28>)

Summary.

1. Qbits can represent multiple states simultaneously.
2. Algorithm by qbits and quantum gates can be much more efficient for some cases.

e.g. Grover search algorithm

Mathematics of Quantum Computing

Recall. 2020 Hyke Hwarang Fall Seminar

- A 'qubit space' is a two-dimensional Hilbert space denoted by $\mathfrak{q}\mathbb{H}$, whose orthonormal basis is $\{|0\rangle, |1\rangle\}$. prob. $|0\rangle$ $|\alpha|^2$
 $|1\rangle$ $|\beta|^2$
- A 'qubit state' $|\psi\rangle \in \mathfrak{q}\mathbb{H}$ can be represented by

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad \alpha, \beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1.$$

- An ' n -qubit' is an element of $\mathfrak{q}\mathbb{H}^{\otimes n} := \bigotimes_{k=1}^n \mathfrak{q}\mathbb{H}$, whose basis is

$$\{|x_1\rangle \otimes \cdots \otimes |x_n\rangle : x_i \in \{0, 1\}, i = 1, \dots, n\},$$

$$|x_1\rangle \otimes \cdots \otimes |x_n\rangle =: |x_1 \cdots x_n\rangle =: \left| \sum_{i=1}^n 2^{n-i} x_i \right\rangle.$$

- A 'quantum n -gate' U is a unitary operate on $\mathfrak{q}\mathbb{H}^{\otimes n}$

quantum 1-gate

Quantum NOT Gate. $X := \sigma_x := \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.

$$\sigma_x|0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle,$$

$$\sigma_x|1\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle.$$

Quantum Gates

quantum 2-gate

Controlled NOT Gate. $\Lambda^1(X) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$

$$\begin{array}{cccc} |00\rangle & |01\rangle & |10\rangle & |11\rangle \\ \text{"} & \text{"} & \text{"} & \text{"} \\ |0\rangle & |1\rangle & |2\rangle & |3\rangle \end{array}$$

$$|00\rangle \rightarrow |00\rangle, \quad |01\rangle \rightarrow |01\rangle, \quad |10\rangle \rightarrow |11\rangle, \quad |11\rangle \rightarrow |10\rangle,$$

$$\Lambda^1(X)|00\rangle = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = |00\rangle,$$

$$\Lambda^1(X)|01\rangle = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = |01\rangle,$$

$$\Lambda^1(X)|10\rangle = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = |11\rangle,$$

$$\Lambda^1(X)|11\rangle = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = |10\rangle$$

Quantum Gates

quantum 2-gate

Swap Gate. $S = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

$$|00\rangle \rightarrow |00\rangle, \quad |01\rangle \rightarrow |10\rangle, \quad |10\rangle \rightarrow |01\rangle, \quad |11\rangle \rightarrow |11\rangle,$$

$$S|00\rangle = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = |00\rangle,$$

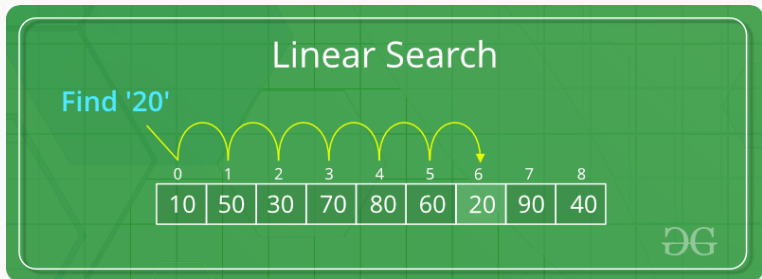
$$S|01\rangle = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = |10\rangle,$$

$$S|10\rangle = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = |01\rangle,$$

$$S|11\rangle = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = |11\rangle,$$

Grover Search Algorithm

Classical Search Algorithm. Complexity: $O(N)$ $\Leftarrow$ Data set of length N is given



<https://www.geeksforgeeks.org/searching-algorithms/>

Note. Grover Search Algorithm can do this in $O(\sqrt{N})$.

Grover Search Algorithm

Problem. Let $S \subset \{0, \dots, N-1 = 2^n - 1\}$ be the set of objects we are searching for, and let $m \geq 1$ be the cardinality of S . We want algorithm searching $x \in S$.

Settings. $\mathbb{H}_S := \text{Span}\{|x\rangle : x \in S\} \subset \mathbb{H}^{\otimes n}$, $\mathbb{H}_{S^\perp} = \text{span}\{|x\rangle : x \in S^\perp\}$
 $P_S := \sum_{x \in S} |x\rangle\langle x|$, $|\Psi_S\rangle := \frac{1}{\sqrt{|S|}} \sum_{x \in S} |x\rangle$. $P_{S^\perp} = \sum_{x \in S^\perp} |x\rangle\langle x|$
 $|\Psi_{S^\perp}\rangle = \frac{1}{\sqrt{|S^\perp|}} \sum_{x \in S^\perp} |x\rangle$

Similarly, $\mathbb{H}_{S^\perp}$, $P_{S^\perp}$, and $|\Psi_{S^\perp}\rangle$ is defined for $S^\perp := \{0, \dots, N-1\} \setminus S$.

Note that every $|\Psi\rangle \in \mathbb{H}^{\otimes n}$ can be represented as

$$|\Psi\rangle = \sum_{x \in S^\perp} \psi_x |x\rangle + \sum_{x \in S} \psi_x |x\rangle, \quad = \sum_{x=0}^{N-1} \psi_x |x\rangle$$

and the probability to observe $x \in S$ when observing $|\Psi\rangle$ is

$$\sum_{x \in S} |\psi_x|^2 = \|P_S |\Psi\rangle\|^2.$$

Grover Search Algorithm

Settings. The 'oracle-function' $g : \{0, \dots, N-1\} \rightarrow \{0, 1\}$ is defined by

$$g(x) := \begin{cases} 0 & x \in S^\perp, \\ 1 & x \in S, \end{cases} = I_S$$

and the 'oracle' $\hat{U}_g$ is a quantum $(n+1)$ -gate on $\mathbb{H}^{\otimes n} \otimes \mathbb{H}$ defined by

$$\hat{U}_g(|x\rangle_n \otimes |y\rangle_1) := |x\rangle \otimes |y + g(x)\rangle = \begin{cases} |x\rangle \otimes |y\rangle, & x \in S^\perp \\ |x\rangle \otimes |1+y\rangle & x \in S \end{cases}$$

$$\begin{aligned} 0+0 &= 0 \\ 0+1 &= 1 \\ 1+0 &= 1 \\ 1+1 &= 0 \end{aligned}$$

Grover Search Algorithm

Lemma

For any $|\Psi\rangle \in \mathbb{H}^{\otimes n}$, one has

$$P_S + P_{S^\perp} = I$$

$$\hat{U}_g(|\Psi\rangle \otimes |-\rangle) = (R_{S^\perp} |\Psi\rangle) \otimes |-\rangle,$$

where $|-\rangle := \frac{|0\rangle - |1\rangle}{\sqrt{2}}$, $R_{S^\perp} := 2P_{S^\perp} - \textcircled{1}$ ↗ identity operator

Proof. $\hat{U}_g(|x\rangle \otimes |0\rangle) = |x\rangle \otimes |0 + g(x)\rangle = \begin{cases} |x\rangle \otimes |0\rangle & x \in S^\perp \\ |x\rangle \otimes |1\rangle & x \in S \end{cases}$

$$\hat{U}_g(|x\rangle \otimes |1\rangle) = |x\rangle \otimes |1 + g(x)\rangle = \begin{cases} |x\rangle \otimes |1\rangle & x \in S^\perp \\ |x\rangle \otimes |0\rangle & x \in S \end{cases}$$

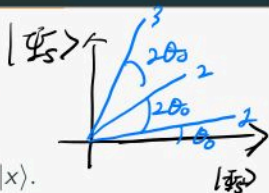
$$\Rightarrow \hat{U}_g(|x\rangle \otimes |-\rangle) = \begin{cases} |x\rangle \otimes |-\rangle & x \in S^\perp \\ -|x\rangle \otimes |-\rangle & x \in S \end{cases} = (-1)^{g(x)} |x\rangle \otimes |-\rangle$$

$$\begin{aligned} \Rightarrow \hat{U}_g(|\Psi\rangle \otimes |-\rangle) &= \left(\underbrace{\sum_{x \in S^\perp} \Psi_x |x\rangle}_{= P_{S^\perp} |\Psi\rangle} - \underbrace{\sum_{x \in S} \Psi_x |x\rangle}_{= P_S |\Psi\rangle} \right) \otimes |-\rangle = (P_{S^\perp} - P_S) |\Psi\rangle \otimes |-\rangle \\ &= (2P_{S^\perp} - I) |\Psi\rangle \otimes |-\rangle \end{aligned}$$

Grover Search Algorithm

Grover iteration operator. $\hat{G} := (R_{\Psi_0} \otimes \mathbf{1}) \hat{U}_g$, where

$$R_{\Psi_0} := 2|\Psi_0\rangle\langle\Psi_0| - \mathbf{1}, \quad \Psi_0 := \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle.$$



Lemma

For $j \in \mathbb{N}$, let $|\hat{\Psi}_j\rangle := (\hat{G})^j |\hat{\Psi}_0\rangle$, where $|\hat{\Psi}_0\rangle := |\Psi_0\rangle \otimes |-\rangle$. Then, we have $|\hat{\Psi}_j\rangle = |\Psi_j\rangle \otimes |-\rangle$, where

$$\begin{aligned} |\Psi_j\rangle &:= \cos \theta_j |\Psi_{S^\perp}\rangle + \sin \theta_j |\Psi_S\rangle, & |\Psi_S\rangle &= \frac{1}{\sqrt{m}} \sum_{x \in S} |x\rangle \\ |\Psi_0\rangle &= \cos \theta_0 |\Psi_{S^\perp}\rangle + \sin \theta_0 |\Psi_S\rangle, & |\Psi_{S^\perp}\rangle &= \frac{1}{\sqrt{N-m}} \sum_{x \in S^\perp} |x\rangle \\ \theta_j &:= (2j+1)\theta_0, & \theta_0 &:= \arcsin\left(\sqrt{\frac{m}{N}}\right) \in \left[0, \frac{\pi}{2}\right]. \end{aligned}$$

n -qubit $\{ \underbrace{\hspace{2cm}}_m \}$

Grover Search Algorithm

Proof. $|\Psi_0\rangle = (\cos\theta_0 |\Psi_{s^\perp}\rangle + \sin\theta_0 |\Psi_s\rangle)$ for $i=0$

suppose $|\hat{\Psi}_j\rangle = (\cos\theta_j |\Psi_{s^\perp}\rangle + \sin\theta_j |\Psi_s\rangle) \otimes |-\rangle = |\Psi_j\rangle \otimes |-\rangle$

$|\hat{\Psi}_{j+1}\rangle = \hat{G} |\hat{\Psi}_j\rangle = (R_{\Psi_0} \otimes I) \hat{U}_g (|\Psi_j\rangle \otimes |-\rangle)$

$= (R_{\Psi_0} \otimes I) ((\cos\theta_j |\Psi_{s^\perp}\rangle - \sin\theta_j |\Psi_s\rangle) \otimes |-\rangle)$ $R_{\Psi_0} = 2|\Psi_0\rangle\langle\Psi_0| - I$

$= R_{\Psi_0} (\cos\theta_j |\Psi_{s^\perp}\rangle - \sin\theta_j |\Psi_s\rangle) \otimes |-\rangle$

$= 2\cos\theta_j |\Psi_0\rangle \cos\theta_0 - 2\sin\theta_j |\Psi_0\rangle \sin\theta_0 - \cos\theta_j |\Psi_{s^\perp}\rangle + \sin\theta_j |\Psi_s\rangle$

$= |\Psi_0\rangle (2\cos\theta_j \cos\theta_0 \sin\theta_0 - 2\sin\theta_j \sin^2\theta_0 + \sin\theta_j)$

$+ |\Psi_{s^\perp}\rangle (\underbrace{2\cos\theta_j \cos\theta_0 \sin\theta_0 - 2\sin\theta_j \sin^2\theta_0 + \sin\theta_j}_{= \cos\theta_j \sin 2\theta_0 + \sin\theta_j (1 - 2\sin^2\theta_0) = \cos\theta_j \sin 2\theta_0 + \sin\theta_j \cos 2\theta_0}$

$= \sin(\theta_j + 2\theta_0) |\Psi_s\rangle + \cos(\theta_j + 2\theta_0) |\Psi_{s^\perp}\rangle$

Grover Search Algorithm

$$\cos \theta_j |\Psi_S\rangle + \sin \theta_j |\Psi_S^\perp\rangle$$

The probability to observe $x \in S$ when observing $|\Psi_j\rangle$ is

$$\|P_S|\Psi_j\rangle\|^2 = \sin^2 \theta_j = \sin^2[(2j+1)\theta_0]$$

what we need to maximize.

$$j = \left\lfloor \frac{\pi}{4\theta_0} \right\rfloor \implies \|P_S|\Psi_j\rangle\|^2 \approx 1.$$

More precisely, since $j \leq \frac{\pi}{4\theta_0} < j+1$, we have

$$-\theta_0 \leq \frac{\pi}{2} - (2j+1)\theta_0 < \theta_0 \implies \|P_S|\Psi_j\rangle\|^2 \geq 1 - \sin^2 \theta_0 = 1 - \frac{m}{N}.$$

$$N = 2^n$$

$$\frac{1}{N} \approx \frac{m}{N} \rightarrow \infty$$

Grover Search Algorithm

$$\theta_0 = \arcsin \sqrt{\frac{m}{N}}$$

One can obtain $|\Psi_j\rangle$ after j -th times of $\hat{G}$.

$$j = \left\lfloor \frac{\pi}{4\theta_0} \right\rfloor = \left\lfloor \frac{\pi}{4 \arcsin \left(\sqrt{\frac{m}{N}} \right)} \right\rfloor = \mathcal{O} \left(\sqrt{\frac{N}{m}} \right)$$

Hence, time complexity of Grover Search Algorithm becomes $\mathcal{O}(\sqrt{N})$.

THANK YOU FOR LISTENING